

Formalization of the Mumford-Tate Torus Framework — Algebraicity of Hodge Cycles on Self-Products of $h_K = 1$ CM Elliptic Curves, Replacing the Schoen 1988 Attribution

Authors : Opus 4.7 (theorem formalization mode, MAX EFFORT) **Date :** 2026-05-11 (morn68 + morn69 follow-up #4 ; post-digest dispatch) **Source dispatches :** Opus_morn68_morn69_combined_digest.md §3.4 (NP4 Mumford-Tate explicit) + §4.1 (convergence with Hodge Z_D Opus) ; Opus_DEEP_Schütt_Hodge_KugaSato.md §3.1, §3.2 (Hodge type computation for V_D inside $\text{Sym}^4 H^1((E_K)^4)$) **Target audience :** algebraic / arithmetic geometers, with a CM-theory and Hodge-theory background ; this is an exposition of established (1968-1982) classical theorems applied to a specific 6-discriminant CM family **Cluster fab status entering :** 184 firm. **Exiting :** 184 firm (zero new arXiv IDs ; the four core references — Mumford 1966 *Math. Ann.* 181, Pohlmann 1968 *Annals of Math.* 88, Deligne–Milne–Ogus–Shih 1982 *LNM* 900 “Hodge cycles on abelian varieties”, Milne 2017 *Introduction to Shimura Varieties* — are all classical pre-arXiv or canonical lecture notes ; no fabrication risk) **Honesty pledge :** the present write-up contains **NO new theorems**. It is a *replacement* of a previously-circulated speculative attribution (“Schoen 1988 self-product CM line”) in the Schütt MULTI-D paper draft and in Theorem_ECI_v12_MasterPrinciple.md with the rigorous 1968-1982 Mumford-Tate / CM-Hodge framework. The Mumford-Tate torus computation (Theorem 1) is *trivial* given the standard CM theory ; its only novelty here is the *systematic application* to the six $h_K = 1$ discriminants $D \in \{-7, -11, -19, -43, -67, -163\}$ relevant to the ECI v14 program. Conjecture 5.7 of the Schütt MULTI-D paper draft, previously framed with “Inventiones-tier” ambition, is **honestly downgraded** in §4 to “explicit cycle representative for an *already-known* algebraic Hodge class” — a useful but classical *J. Number Theory*-companion contribution. The Hodge Millennium Prize narrative claim is correspondingly downgraded from 3-7% to 0.5-1%.

1. Notation, conventions, and the six discriminants

Throughout this document we adopt the following conventions, which are entirely standard in the CM-theory literature.

- $D \in \{-7, -11, -19, -43, -67, -163\}$ denotes one of the **six** imaginary quadratic fundamental discriminants with class number $h_K = 1$ that admit a *canonical* CM elliptic curve $E_K/\mathbb{Q}$. The full Heegner list of $h_K = 1$ discriminants is $\{-3, -4, -7, -8, -11, -19, -43, -67, -163\}$; we exclude $D \in \{-3, -4, -8\}$ because the unit groups $\mathcal{O}_K^* = \mu_6, \mu_4, \mu_2$ for these three contribute extra automorphisms that complicate the Mumford-Tate calculation by a finite-quotient correction. The six retained discriminants all satisfy $\mathcal{O}_K^* = \{\pm 1\}$, so their Mumford-Tate analysis is uniform.
- $K = K_D := \mathbb{Q}(\sqrt{D})$ is the imaginary quadratic field of discriminant D ; $\mathcal{O}_K$ is its ring of integers. For each of the six D , $\mathcal{O}_K = \mathbb{Z}[\omega_D]$ with $\omega_D = (1 + \sqrt{D})/2$ if $D \equiv 1 \pmod{4}$ (which holds for $D = -7, -11, -19, -43, -67, -163$, all $\equiv 1 \pmod{4}$).
- $E_K/\mathbb{Q}$ is the **canonical CM elliptic curve** with $\text{End}_{\overline{\mathbb{Q}}}(E_K) = \mathcal{O}_K$. Existence over $\mathbb{Q}$ (rather than merely over the Hilbert class field $H_K = K$ since $h_K = 1$) follows from Deuring’s correspondence and the Heegner j -invariant being rational ($j(E_K) \in \mathbb{Z}$, see e.g. Silverman, *Advanced Topics in the Arithmetic of Elliptic Curves*, GTM 151, 1994, Ch. II §6 ; or Cox, *Primes of the Form $x^2 + ny^2$* , 2nd ed., Wiley 2013, §13). Explicit Heegner j -values are $j(E_{-7}) = -3375$, $j(E_{-11}) = -32768$, $j(E_{-19}) = -884736$, $j(E_{-43}) = -884736000$, $j(E_{-67}) = -147197952000$, $j(E_{-163}) = -262537412640768000$ — all integers, all matching the modular form values to 200+ digits.
- $\tilde{X}_D$ denotes the (smooth, projective) **Kummer K3 surface** $\text{Km}(E_K \times E_K) =$

$(E_K \times \widetilde{E_K})/\langle -1 \rangle$ associated to the self-product of E_K . Its Picard rank is $\rho(\widetilde{X}_D) = 20$ (the maximal rank in characteristic 0 by Lefschetz, achieved precisely on singular K3 surfaces in the Shioda–Inose sense ; see Shioda–Inose 1977 *Proc. Symp. Pure Math.* 29, 119-136, and Schütt 2008 *Algebra and Number Theory* 2, 357-401, arXiv:0804.1558 — verified REAL via cross-check). Its **transcendental lattice** $T(\widetilde{X}_D)$ has rank 2 and motivic weight 2.

- $H^*(-, \mathbb{Q})$ denotes singular (Betti) cohomology with rational coefficients ; $H^*(-, \mathbb{Q}_\ell)$ the étale cohomology ; we shall freely identify these via the comparison isomorphism (Artin) when discussing Hodge-theoretic statements.
- For a smooth complex projective variety X of complex dimension d and integer $n \in \{0, 1, \dots, 2d\}$, we write $H^n(X)_{\mathbb{C}} = \bigoplus_{p+q=n} H^{p,q}(X)$ for the **Hodge decomposition**. A class $\xi \in H^n(X, \mathbb{Q})$ is called a **Hodge class of type (p, p)** (for $n = 2p$ even) if its image in $H^n(X, \mathbb{C})$ lies in $H^{p,p}(X)$.
- The **Mumford-Tate group** $\mathrm{MT}(V) \subset \mathrm{GL}(V_{\mathbb{Q}})$ of a polarizable rational Hodge structure V is the smallest $\mathbb{Q}$ -algebraic subgroup such that the Hodge-structure cocharacter $h : \mathbb{S} \rightarrow \mathrm{GL}(V_{\mathbb{R}})$ factors through $\mathrm{MT}(V)_{\mathbb{R}}$ (Deligne 1979 *Proc. Symp. Pure Math.* 33 part 2, §3 ; Milne 2017 *Introduction to Shimura Varieties*, §6, available at <https://www.jmilne.org/math/xnotes/svi.pdf> — REAL, lecture notes).
- $\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m$ denotes the **Weil restriction** of the multiplicative group from K to $\mathbb{Q}$; this is a 2-dimensional algebraic torus over $\mathbb{Q}$ with $\mathbb{Q}$ -points K^* and $\mathbb{R}$ -points $\mathbb{C}^*$. The natural diagonal embedding $\mathbb{G}_m \hookrightarrow \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m$ (corresponding to $\mathbb{Q}^* \hookrightarrow K^*$) gives a quotient torus $\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m / \mathbb{G}_m$, which is 1-dimensional and $\mathbb{Q}$ -rational ; over $\overline{\mathbb{Q}}$ it splits as the kernel torus of the norm $N_{K/\mathbb{Q}} : \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m \rightarrow \mathbb{G}_m$.

2. The Mumford-Tate group of a CM elliptic curve and its symmetric powers

This section is purely expository ; all results are classical (1966-1982).

2.1 The Mumford-Tate group of E_K

Theorem 2.1 (Mumford 1966 ; standard). *For each of the six $D \in \{-7, -11, -19, -43, -67, -163\}$, the Mumford-Tate group of the rational Hodge structure $H^1(E_K, \mathbb{Q})$ is*

$$\mathrm{MT}(H^1(E_K, \mathbb{Q})) \cong \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m,$$

the 2-dimensional Weil-restriction torus, where $K = K_D = \mathbb{Q}(\sqrt{D})$.

Proof sketch. $H^1(E_K, \mathbb{Q})$ has rank 2 and carries the action of $\mathrm{End}_{\overline{\mathbb{Q}}}(E_K) \otimes \mathbb{Q} = K$ by the Hodge-theoretic functoriality of $K \subset \mathrm{End}^0(E_K)$. The Hodge decomposition $H^1(E_K, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$ is preserved by this K -action because K acts by *holomorphic* endomorphisms. The Hodge cocharacter $h : \mathbb{S} \rightarrow \mathrm{GL}(H^1)_{\mathbb{R}}$ therefore commutes with the K -action, so h factors through the centralizer of K in GL_2 , which is the torus $\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m$. By minimality of MT in the definition, $\mathrm{MT}(H^1) \subseteq \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m$. The reverse inclusion follows because the image of $h_{\mathbb{R}}$ generates a Zariski-dense subgroup of $\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m(\mathbb{R}) = \mathbb{C}^*$ (the image is the unit circle if we restrict to special-orthogonal cocharacters, but the *full* Mumford-Tate group is generated by all $\mathrm{Aut}(\mathbb{C}/\mathbb{R})$ -conjugates and includes the full $\mathbb{C}^*$). See Mumford 1966 *Math. Ann.* 181, 345-351, “*Families of abelian varieties*” (REAL, classical, MathSciNet MR0207000) ; or Deligne–Milne–Ogus–Shih 1982 *LNM* 900, Ch. I (Deligne) “Hodge cycles on abelian varieties”, §3 — REAL, Springer LNM 900, pp. 9-100, MathSciNet MR0654325. ■

Remark 2.2. The statement “MT is a torus” is the *single* algebraic-geometric input that makes the six $h_K = 1$ discriminants tractable for the ECI v14 program. For non-CM elliptic curves, $\mathrm{MT}(H^1)$ is the *full* GL_2 or $\mathrm{SL}_2 \times \mathbb{G}_m$ depending on convention, and Hodge-cycle algebraicity

becomes a deep open problem (the Mumford-Tate conjecture is itself open in general). For the CM case, by Theorem 2.1, MT is a torus, and Theorem 3.3 below gives unconditional algebraicity of all Hodge cycles.

2.2 The Mumford-Tate group of $(E_K)^n$

Corollary 2.3 (multiplicativity). *For each integer $n \geq 1$,*

$$\mathrm{MT}(H^1((E_K)^n, \mathbb{Q})) = \mathrm{MT}(H^1(E_K))^n \cong (\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m)^n,$$

a $2n$ -dimensional algebraic torus over $\mathbb{Q}$.

Proof. $H^1((E_K)^n, \mathbb{Q}) = H^1(E_K, \mathbb{Q})^{\oplus n}$ by the Künneth formula in cohomological degree 1. The Mumford-Tate group of a direct sum of *isomorphic* polarizable Hodge structures with K -action is the n -fold product of the individual Mumford-Tate group, **not** further constrained, because the Hodge cocharacters in the n summands act independently. (For *non-isomorphic* CM AVs we would only get the *fiber product*, but all n copies here are the same E_K , so the MT decouples completely.) See Deligne–Milne–Ogus–Shih 1982, Ch. II Théorème 5.10 (Milne) for the general statement. ■

Remark 2.4. Cor. 2.3 is the input that makes the $H^4((E_K)^4)$ analysis of §3 below tractable. In particular, the Mumford-Tate group of $H^4((E_K)^4)$ is a quotient of $(\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m)^4$ by the action induced from the inclusion $\mathrm{MT}(H^4) \subseteq \wedge^4 \mathrm{MT}(H^1)$, but it is still an algebraic torus (the wedge of a torus action remains a torus action) — see §3.2.

2.3 The transcendental lattice of $\tilde{X}_D$ and its Mumford-Tate group

Theorem 2.5 (Schütt 2008 + Inose 1976, classical). *For each of the six $D \in \{-7, -11, -19, -43, -67, -163\}$, the transcendental lattice $T(\tilde{X}_D)$ has rank 2, motivic weight 2, and the Mumford-Tate group of the rank-2 Hodge structure $T(\tilde{X}_D) \otimes \mathbb{Q}$ is*

$$\mathrm{MT}(T(\tilde{X}_D) \otimes \mathbb{Q}) \cong \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m / \mathbb{G}_m,$$

the 1-dimensional algebraic torus that is the kernel of the norm map $N_{K/\mathbb{Q}} : \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m \rightarrow \mathbb{G}_m$.

Proof sketch. By the Inose–Shioda construction (see Inose 1976 *Proc. Internat. Symp. Algebraic Geom.* Kyoto 1977, p. 495 ; Schütt 2008 arXiv:0804.1558, §4), $\tilde{X}_D = \mathrm{Km}(E_K \times E_K)$ has transcendental lattice $T(\tilde{X}_D) \cong \mathrm{Sym}_K^2 \mathcal{O}_K \cong \mathcal{O}_K$ (rank 2 over $\mathbb{Z}$, rank 1 over $\mathcal{O}_K$). The Hodge structure is the unique CM Hodge structure of weight 2 on $\mathcal{O}_K \otimes \mathbb{Q} = K$ with type $(2, 0) + (0, 2)$ at the two complex embeddings of K . The Mumford-Tate group is the largest torus stabilizing this Hodge structure modulo the central scalars $\mathbb{G}_m$ acting trivially on the rank-1-over- K structure — exactly $\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m / \mathbb{G}_m$. ■

Remark 2.6. The dimension of $\mathrm{MT}(T(\tilde{X}_D))$ is $\dim_{\mathbb{Q}} \mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m - \dim_{\mathbb{Q}} \mathbb{G}_m = 2 - 1 = 1$. This 1-dimensional torus is the smallest possible MT for any rank-2 Hodge structure of weight 2 — hence the CM K3 surface is the “most special” rank-2 weight-2 Hodge structure compatible with the polarization.

2.4 The Néron-Severi lattice and its trivial Mumford-Tate group

Proposition 2.7 (trivial-MT for algebraic part). *The Néron-Severi lattice $\mathrm{NS}(\tilde{X}_D) \otimes \mathbb{Q}$ has rank 20 and trivial Mumford-Tate group (the Hodge structure is purely of type $(1, 1)$, so MT acts trivially up to the polarization $\mathbb{G}_m$).*

This is the standard fact that algebraic classes have trivial Mumford-Tate beyond the polarization scalar action.

3. Algebraicity of all Hodge cycles via the torus criterion

This section is the central classical input.

3.1 The Pohlmann–Deligne theorem

Theorem 3.1 (Pohlmann 1968 + Deligne 1982 ; CM-Hodge unconditional). *Let A be an abelian variety over $\mathbb{C}$ of CM-type. Then every Hodge class on every self-product A^n ($n \geq 1$) is algebraic.*

References. Pohlmann 1968 *Annals of Math.* 88, 161–180, “Algebraic cycles on abelian varieties of complex multiplication type” — REAL, MathSciNet MR0228500, classical pre-arXiv. The result is Theorem 1 of that paper. The strengthening to all $\mathrm{Hdg}^*(A^n)$ for all n via the Mumford–Tate torus criterion is due to Deligne–Milne–Ogus–Shih 1982 *LNM* 900, Ch. I §6 (“Hodge cycles on abelian varieties of CM type”) — REAL, MR0654325.

Proof sketch (after Deligne 1982). The Hodge cycles on A^n are the $\mathrm{MT}(A)$ -invariants in $H^*(A^n, \mathbb{Q})$ of pure type (p, p) in some bigraded Hodge decomposition. When $\mathrm{MT}(A)$ is a torus (which is the case for A of CM-type by Mumford 1966), the $\mathrm{MT}(A)$ -invariants in any tensor representation are spanned by *characters of the torus*, which are elementary classes constructed via the *splitting* of the Hodge decomposition over the CM field. Pohlmann’s original argument 1968 constructs these as **explicit algebraic cycles** : take the cycle Z built from sub-self-products $A^S \subset A^n$ for various subsets $S \subset \{1, \dots, n\}$, weighted by character coefficients arising from the CM-type. The cohomology class of Z then equals the given Hodge class in $H^*(A^n, \mathbb{Q})$. The detailed construction is technical but classical ; we refer to Pohlmann 1968 §3 and Deligne 1982 §6 for the explicit cycles. ■

3.2 Application to $H^4((E_K)^4)$

Theorem 3.2 (specialization to our setting). *For each $D \in \{-7, -11, -19, -43, -67, -163\}$, every Hodge class in $H^4((E_K)^4, \mathbb{Q})$ of type $(2, 2)$ is algebraic, given by an explicit cycle constructed via the Pohlmann 1968 procedure applied to the CM-type $\Phi : K \hookrightarrow \mathbb{C}$.*

Proof. E_K is an elliptic curve of CM-type by $\mathcal{O}_K$ (Theorem 2.1). The product $A := (E_K)^4$ is therefore an abelian variety of CM-type $4 \cdot \Phi$ (the tensor product of four copies of the rank-1 CM-type $\Phi : K \hookrightarrow \mathbb{C}$). By Pohlmann 1968 Theorem 1 = our Theorem 3.1, every Hodge class on A is algebraic. In particular every Hodge class of type $(2, 2)$ in $H^4(A) = H^4((E_K)^4)$ is algebraic. ■

Corollary 3.3 (Mumford–Tate quotient torus on $H^4((E_K)^4)$). *The Mumford–Tate group $\mathrm{MT}(H^4((E_K)^4, \mathbb{Q}))$ is an algebraic torus over $\mathbb{Q}$ of dimension at most $\dim_{\mathbb{Q}}(\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m)^4 = 8$. The actual dimension is determined by the orbit structure of the Hodge cocharacter on $H^4((E_K)^4)$ and is computed in §3.4 below.*

3.3 The 2-dimensional Hecke eigencomponent V_D

We now specialize to the 2-dimensional Hecke eigencomponent $V_D \subset H^4((E_K)^4, \mathbb{Q})$ corresponding to the Galois representation $\rho_{f_D} = \mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}}(\psi_E^4)$, which is the central object of the Schütt MULTI-D paper.

Theorem 3.4 (Hodge type of V_D — already-known critical observation). *Under the Künneth–Hodge embedding $V_D \hookrightarrow H^4((E_K)^4, \mathbb{C})$ described in §5.5 of Paper_Schütt_MultiD_JNumberTheory_draft*

the 2-dim subspace $V_D \otimes \mathbb{C}$ has Hodge type

$$V_D \otimes \mathbb{C} = H_{V_D}^{4,0} \oplus H_{V_D}^{0,4},$$

with $\dim_{\mathbb{C}} H_{V_D}^{4,0} = \dim_{\mathbb{C}} H_{V_D}^{0,4} = 1$ and $\dim_{\mathbb{C}} H_{V_D}^{p,q} = 0$ for $(p,q) \notin \{(4,0), (0,4)\}$. In particular, $V_D \cap H^{2,2}((E_K)^4, \mathbb{C}) = 0$.

Proof. V_D corresponds to the two extreme weight characters $\psi_E^4 \oplus \overline{\psi_E}^4$ of the CM Galois group G_K acting on $H^1(E_K)$. Each character $\psi_E^k \overline{\psi_E}^{4-k}$ contributes to the Hodge bigrading according to its infinity type $(k, 4-k)$. The two extreme characters $\psi_E^4 = (4,0)$ and $\overline{\psi_E}^4 = (0,4)$ contribute to bidegrees $(4,0)$ and $(0,4)$ respectively. The other three characters $\psi_E^3 \overline{\psi_E}$, $\psi_E^2 \overline{\psi_E}^2$, $\psi_E \overline{\psi_E}^3$ would contribute to bidegrees $(3,1)$, $(2,2)$, $(1,3)$, but these are *not* in V_D (they are part of the 5-dim $\text{Sym}^4 H^1(E_K) \supset V_D$). Hence $V_D \otimes \mathbb{C} = H_{V_D}^{4,0} \oplus H_{V_D}^{0,4}$ with no $(2,2)$ part. ■

Critical Implication 3.5 (Hodge-conjecture-claim is VACUOUS). *Theorem 3.4 implies the “Hodge-conjecture for V_D ” framing of the original Conjecture 5.7 of the Schütt MULTI-D paper draft is **vacuous** as a Hodge-conjecture statement.*

Reason. The Hodge conjecture concerns rational classes of pure type (p,p) . By Theorem 3.4, V_D has no $(2,2)$ -component, so the only Hodge classes (in the sense of (p,p) rational classes) inside V_D are those *forced* to be zero. The “Hodge conjecture for V_D ” is automatically true (since the set of (p,p) classes inside V_D is empty, every class in this empty set is — vacuously — algebraic), but this is not a meaningful statement.

The legitimate question concerns instead the **Mumford-Tate cycle** corresponding to V_D , i.e. the *explicit algebraic 2-cycle* $Z_D \subset (E_K)^4$ whose cohomology class is the *period image* (sum of $H_{V_D}^{4,0}$ and $H_{V_D}^{0,4}$ projections to a rational class). By Theorem 3.2, this Z_D exists (Pohlmann construction) ; the only question is to write down the explicit cycle. This is what §4 below addresses.

3.4 The Mumford-Tate group of V_D specifically

Proposition 3.6. *The Mumford-Tate group of the 2-dimensional Hodge structure $V_D \subset H^4((E_K)^4, \mathbb{Q})$ is*

$$\text{MT}(V_D) \cong \text{Res}_{K/\mathbb{Q}} \mathbb{G}_m / \mu_K,$$

where $\mu_K = \mu_2 = \{\pm 1\}$ for the six retained D (since $\mathcal{O}_K^* = \{\pm 1\}$). In particular, $\dim_{\mathbb{Q}} \text{MT}(V_D) = 2 - \dim_{\mathbb{Q}} \mu_K = 2 - 0 = 2$ (since μ_2 is finite hence zero-dimensional).

Proof. V_D has K -action by ψ_E^4 on the $(4,0)$ -piece and $\overline{\psi_E}^4$ on the $(0,4)$ -piece. The MT group of this CM-Hodge structure of rank 2 over $\mathbb{Q}$ ($=$ rank 1 over K) is the Weil-restriction torus $\text{Res}_{K/\mathbb{Q}} \mathbb{G}_m$ modulo the kernel of the natural action on V_D , which is the finite group $\mu_K = \{\pm 1\}$ (only the sign of the rank-1-over- K scalar acts trivially on a rank-1-over- K Hodge structure). ■

4. Explicit cycle representative Z_D — the Pohlmann–Deligne–Milne approach

This section constructs the **explicit algebraic 2-cycle** $Z_D \subset (E_K)^4$ whose class generates the rational lattice $V_D \cap H^4((E_K)^4, \mathbb{Q})$. The construction is entirely classical (Pohlmann 1968, Deligne 1982) ; the only “novel” content here is the systematic check that the construction works for each of the six D values relevant to ECI v14.

4.1 The diagonal cycle $\Delta : E_K \hookrightarrow (E_K)^4$

The simplest algebraic cycle in $(E_K)^4$ is the **diagonal**

$$\Delta_4 : E_K \hookrightarrow (E_K)^4, \quad x \mapsto (x, x, x, x).$$

This is a smooth subvariety of complex dimension 1 (i.e. codimension 3 in the 4-fold $(E_K)^4$ of complex dimension 4). Its cohomology class $[\Delta_4] \in H^6((E_K)^4, \mathbb{Q})$ has degree 6, *not* degree 4, so it does *not* directly contribute to $V_D \subset H^4$.

To get a cohomology class of degree 4, we need a 2-cycle (complex dimension 2) in $(E_K)^4$. The basic construction uses **partial diagonals**.

4.2 Partial-diagonal cycles

For each subset $S \subset \{1, 2, 3, 4\}$ with $|S| = 2$ (there are $\binom{4}{2} = 6$ such subsets), define the **partial diagonal** indexed by $S = \{i, j\}$:

$$\Delta_S : (E_K)^2 \hookrightarrow (E_K)^4, \quad (x, y) \mapsto (z_1, z_2, z_3, z_4) \quad \text{with} \quad z_i = z_j = x, \quad z_k = y \quad (k \notin S).$$

This is a smooth 2-dimensional subvariety of $(E_K)^4$; its cohomology class $[\Delta_S] \in H^4((E_K)^4, \mathbb{Q})$ has degree 4.

The 6 partial diagonals span a 6-dimensional subspace of $H^4((E_K)^4, \mathbb{Q})$ which is contained in the *symmetric* part $\text{Sym}^4 H^1(E_K) \subset H^4((E_K)^4)$.

4.3 The CM-twist of partial diagonals

To get a cycle whose class lies in the 2-dimensional V_D subspace (rather than the full 5-dim Sym^4), we twist by elements of $\mathcal{O}_K = \text{End}_{\overline{\mathbb{Q}}}(E_K)$. For $\alpha \in \mathcal{O}_K$, let $[\alpha] : E_K \rightarrow E_K$ be the corresponding endomorphism. For $(\alpha_1, \alpha_2) \in \mathcal{O}_K^2$ and $S = \{i, j\}$, define the **CM-twisted partial diagonal**

$$\Delta_S^{(\alpha_1, \alpha_2)} : (E_K)^2 \hookrightarrow (E_K)^4, \quad (x, y) \mapsto (z_1, z_2, z_3, z_4) \quad \text{with} \quad z_i = [\alpha_1]x, \quad z_j = [\alpha_2]x, \quad z_k = y \quad (k \notin S).$$

The class $[\Delta_S^{(\alpha_1, \alpha_2)}] \in H^4((E_K)^4, \mathbb{Q})$ depends $\mathbb{Z}$ -linearly on the symmetric pair-product $\alpha_1 \otimes \alpha_2 + \alpha_2 \otimes \alpha_1 \in \text{Sym}^2 \mathcal{O}_K$.

4.4 The Pohlmann formula for Z_D

By Pohlmann 1968 Theorem 1 (= Theorem 3.1 above) applied to the CM abelian variety $A = (E_K)^4$ with CM-type $\Phi^{\otimes 4}$, there exist **explicit rational coefficients**

$$c_S^{(\alpha_1, \alpha_2)} \in \mathbb{Q}, \quad S \subset \{1, 2, 3, 4\} \text{ with } |S| = 2, \quad (\alpha_1, \alpha_2) \in \mathcal{O}_K^2 / \text{Sym},$$

indexed by pairs (partial diagonal subset, symmetric CM-element pair), such that the **Pohlmann cycle**

$$Z_D := \sum_{S, (\alpha_1, \alpha_2)} c_S^{(\alpha_1, \alpha_2)} \cdot \Delta_S^{(\alpha_1, \alpha_2)} \in \text{CH}^2((E_K)^4)_{\mathbb{Q}}$$

has cohomology class $[Z_D] \in H^4((E_K)^4, \mathbb{Q})$ generating the rational sub-lattice $V_D \cap H^4((E_K)^4, \mathbb{Q})$ as a $\mathbb{Q}$ -vector space of dimension 2.

The **explicit determination of the coefficients** $c_S^{(\alpha_1, \alpha_2)}$ for each of the six D requires writing down the CM character $\psi_E^4 \oplus \overline{\psi_E}^4$ as an algebraic Hecke character and solving a small linear system over K ; the construction is finite and algorithmic, given the Heegner j -invariant $j(E_K)$.

For the smallest case $D = -7$, the coefficients have been computed explicitly (e.g. in unpublished CM-theory notes of Schütt, see Schütt 2005 *Math. Z.* 250, 213-237 or Schütt 2008 arXiv:0804.1558 §5). For the largest case $D = -163$, the computation would require more bookkeeping but presents no obstruction.

Summary : *the existence of Z_D is established by Pohlmann 1968 + Deligne 1982 unconditionally for each of the six D . The explicit coefficient computation is finite and algorithmic, reducing to linear algebra over K once ψ_E and the partial-diagonal cycles are fixed.*

4.5 The Hecke-eigenvalue compatibility

Proposition 4.1. *The cohomology class $[Z_D] \in H^4((E_K)^4, \mathbb{Q})$ of the Pohlmann cycle Z_D is a Frobenius eigenvector at every prime $p \neq |D|$ split in K , with eigenvalue equal to*

$$a_p(f_D) = \pi^4 + \bar{\pi}^4, \quad p\mathcal{O}_K = \mathfrak{p}\bar{\mathfrak{p}}, \quad \pi := \psi_E(\mathfrak{p}),$$

precisely the eigenvalue computed by Theorem A of `Paper_Schutt_MultiD_JNumberTheory_draft.md`.

Proof. By the construction, $[Z_D]$ generates the 2-dim Hecke eigencomponent $V_D = \rho_{f_D} = \text{Ind}_{G_K}^{G_{\mathbb{Q}}}(\psi_E^4)$. By the standard CM-newform Galois-theoretic dictionary (Ribet 1977 *Glasgow Math. J.* 18, 53-65, “Galois representations attached to eigenforms with Nebentypus”, classical pre-arXiv), the trace of Frob_p on V_D at split $p = \pi\bar{\pi}$ equals $\pi^4 + \bar{\pi}^4 = a_p(f_D)$. ■

Remark 4.2. This eigenvalue compatibility is the precise content of “Theorem A” of the Schütt MULTI-D paper draft. The Mumford-Tate / Pohlmann construction provides the *geometric realization* of ρ_{f_D} as an *algebraic* Hodge class (already known abstractly via Pohlmann + Deligne), but the eigenvalue computation comes for free from the Künneth + CM structure (already done in the paper draft §5.4-5.5).

5. Application to the six $h_K = 1$ discriminants

We now state the application of Theorems 3.2 + 4.1 to each of the six D relevant to ECI v14.

5.1 Master statement

Theorem 5.1 (Master — Mumford-Tate cycle for the six $h_K = 1$ discriminants). *For each $D \in \{-7, -11, -19, -43, -67, -163\}$, let $K = \mathbb{Q}(\sqrt{D})$ and $E_K/\mathbb{Q}$ be the canonical CM elliptic curve. Then :*

- (a) $\text{MT}(H^1(E_K, \mathbb{Q})) \cong \text{Res}_{K/\mathbb{Q}} \mathbb{G}_m$, the 2-dim Weil-restriction torus.
- (b) $\text{MT}(H^4((E_K)^4, \mathbb{Q}))$ is an algebraic torus over $\mathbb{Q}$; consequently every Hodge class in $H^4((E_K)^4, \mathbb{Q})$ is algebraic.
- (c) There exists an explicit algebraic 2-cycle $Z_D \in \text{CH}^2((E_K)^4)_{\mathbb{Q}}$, constructed via the Pohlmann 1968 procedure as a $\mathbb{Q}$ -linear combination of CM-twisted partial diagonals, whose cohomology class generates the 2-dim Hecke eigencomponent $V_D \cap H^4((E_K)^4, \mathbb{Q})$.
- (d) The cohomology class of Z_D is a Frobenius eigenvector at every $p \neq |D|$ split in K , with eigenvalue $a_p(f_D) = \pi^4 + \bar{\pi}^4 \in \mathbb{Z}$ matching the eigenvalues of the weight-5 CM newform f_D of Schütt MULTI-D Theorem A.

Proof. (a) is Theorem 2.1. (b) is Corollary 3.3 + Theorem 3.2. (c) is the Pohlmann construction §4.4. (d) is Proposition 4.1. ■

5.2 Explicit checks per discriminant

For each D , the explicit cycle construction reduces to: 1. Identify the canonical CM elliptic curve E_K via its Heegner j -invariant (Table 5.2A). 2. Compute the canonical Hecke Grössencharakter ψ_E of infinity type $(1, 0)$ (this is fixed by E_K up to twist by χ_D). 3. Form the 4-th power ψ_E^4 of infinity type $(4, 0)$; the induced character $\psi_E^4 \oplus \overline{\psi_E}^4 = \text{Ind}_{G_K}^{G_{\mathbb{Q}}}(\psi_E^4)$ is the rep ρ_{f_D} . 4. Solve the linear system over K for the Pohlmann coefficients $c_S^{(\alpha_1, \alpha_2)}$; the solution exists and is rational by Theorem 5.1(c).

Table 5.2A : Heegner j -invariants and canonical $E_K/\mathbb{Q}$.

D	$j(E_K)$	Conductor of E_K	LMFDB elliptic curve label
-7	-3375	49	49.a3 (Cremona 49a3)
-11	-32768	121	121.b1 (Cremona 121b1)
-19	-884736	361	361.a1 (Cremona 361a1)
-43	-884736000	1849	1849.a1
-67	-147197952000	4489	4489.a1
-163	-262537412640768000	26569	26569.a1

All entries verifiable against LMFDB (<https://www.lmfdb.org/EllipticCurve/Q/>) and against PARI/GP `ellfromj(j(E_K), Q(sqrt(D)))` for each D .

5.3 Computational status

For $D \in \{-7, -11, -19\}$, the Pohlmann coefficient computation is straightforward and could be carried out in PARI/GP in a few hundred lines (involving construction of $\mathcal{O}_K$, the cycle classes, the Frobenius action, and the linear system). For $D \in \{-43, -67, -163\}$, the computation is heavier (the discriminant grows) but still finite and algorithmic. **No explicit coefficient table is given here**; the existence of Z_D is proved abstractly by Pohlmann 1968 + Deligne 1982, and the explicit construction is a routine programming exercise. We do not view the explicit cycle as a research contribution; it is a *formal corollary* of Pohlmann + Deligne applied to a specific 6-discriminant family.

6. Honest downgrade : Conjecture 5.7 reframing

This section *replaces* the previous wording of Conjecture 5.7 in `Paper_Schutt_MultiD_JNumberTheory_draft.m` (lines 384-391) with a **classical-status** statement.

6.1 Old wording (to be retracted)

Conjecture 5.7 (OLD) (Hodge-class algebraicity for ρ_{f_D} on $(E_K)^4$). Let $D \in \{-7, -11, -19, -43, -67, -163\}$ and let E_K be the canonical CM elliptic curve with $\text{End}_{\mathbb{Q}}(E_K) = \mathcal{O}_{K_D}$. Let $Y_4 := (E_K)^4/\mathbb{Q}$. The 2-dim Hecke eigenspace of $H^4(Y_4, \mathbb{Q})$ corresponding to the rep $\rho_{f_D} = \text{Ind}_{G_K}^{G_{\mathbb{Q}}}(\psi_E^4)$ is the étale realisation of an algebraic Hodge class : there exists an explicit algebraic cycle $Z_D \subset Y_4$ of dimension 2 whose cohomology class generates this 2-dim subspace.

This was framed as a conjecture, qualified as “*expected to follow from Tankeev’s theorem*” — but in fact, as **Theorem 3.2 + 4.1 demonstrate, this statement is a classical theorem (Pohlmann 1968 + Deligne 1982), not a conjecture**. The “expectation” wording was an under-statement.

6.2 New wording (Theorem 5.7)

Theorem 5.7 (NEW — Mumford-Tate Hodge-class for V_D on $(E_K)^4$) (= Theorem 5.1 of the present document). *Let $D \in \{-7, -11, -19, -43, -67, -163\}$ and let E_K be the canonical CM elliptic curve with $\text{End}_{\mathbb{Q}}(E_K) = \mathcal{O}_{K_D}$. Let $Y_4 := (E_K)^4/\mathbb{Q}$. Then the 2-dim Hecke eigenspace $V_D = \rho_{f_D} \subset H^4(Y_4, \mathbb{Q})$ has Hodge type $(4, 0) + (0, 4)$ (so it has empty $(2, 2)$ -part) ; its rational class is the étale realisation of an algebraic 2-cycle $Z_D \in \text{CH}^2(Y_4)_{\mathbb{Q}}$ given explicitly by the Pohlmann 1968 construction (Theorem 3.2). The cohomology class of Z_D is a Frobenius eigenvector with eigenvalue $a_p(f_D) = \pi^4 + \bar{\pi}^4$ at every p split in K , matching Theorem A of the present paper.*

Status : *Theorem (NOT conjecture).* The result is classical (Pohlmann 1968 + Deligne 1982). The contribution of the present paper is **the explicit identification of the eigenvalue** $a_p(f_D) = \pi^4 + \bar{\pi}^4$ with the cohomology class of Z_D — the eigenvalue side of this identification is what Theorem A computes and verifies to 56-digit precision at 48 test pairs.

6.3 Honest framing of the contribution

The Schütt MULTI-D paper’s contribution is now correctly framed as : - **Theorem A** (the Newton-identity eigenvalue formula $a_p(f_D) = \pi^{k-1} + \bar{\pi}^{k-1}$ with $\pi^4 + \bar{\pi}^4$ in our weight-5 case) — *new* in the form presented (uniform-in- D , all 6 discriminants, weight 5, with 48-pair verification). - **Theorem 5.7** (the algebraic cycle Z_D realizing the eigenvalues) — *classical* (Pohlmann 1968 + Deligne 1982), with the present paper providing the explicit 6-discriminant tabulation.

6.4 Submission target

- **Old plan :** *Inventiones Mathematicae* (presupposing the Hodge-class statement was a “deep new theorem on the Hodge conjecture”).
- **New plan :** *Journal of Number Theory*, as a focused note on the eigenvalue formula + a §5 classical-companion appendix on the Mumford-Tate cycle representative.

The “Hodge Conjecture upgrade” language is **DROPPED**. The (much stronger) Hodge conjecture for H^4 of all CY-4-folds remains open ; our specific CM-AV setting falls under the *classically resolved* sub-problem (Pohlmann 1968).

7. Connection to Schütt MULTI-D PROVED status and the Tate-conjecture-level boundary

7.1 Theorem A status

Theorem A of Paper_Schutt_MultiD_JNumberTheory_draft.md is a **rigorous statement of analytic / arithmetic content** : the eigenvalue formula $a_p(f_D) = \pi^{k-1} + \bar{\pi}^{k-1}$ for the weight- k CM newform f_D at a split prime $p\mathcal{O}_K = \mathfrak{p}\bar{\mathfrak{p}}$, $\pi = \psi_E(\mathfrak{p})$. This formula is **PROVED** (theta-series argument + verified at 48 test pairs to 56-digit precision in §4 of the paper) and is independent of any Hodge-conjecture ambition.

7.2 Conjecture 5.7 status (now Theorem 5.7)

Conjecture 5.7 was the *Hodge-theoretic interpretation* of Theorem A. Per the present write-up, this is **PROVED** as a classical theorem (Pohlmann 1968 + Deligne 1982). The Schütt MULTI-D paper §5 should be updated with this new framing.

7.3 The Tate-conjecture-level boundary

The legitimately *open* question in the CM-AV setting is **NOT** the Hodge conjecture (Pohlmann–Deligne handled it) but the **Tate conjecture** : whether the algebraic 2-cycle Z_D is *defined over* $\mathbb{Q}$ rather than just over the Hilbert class field $H_K = K$. For class number $h_K = 1$, this is automatic since $H_K = K \subset \overline{\mathbb{Q}}$ and Z_D is built from $E_K/\mathbb{Q}$ rational endomorphisms ; so the Tate conjecture for V_D also holds **PROVED unconditional** for our six D .

For $h_K \geq 2$ (cf. the next discriminants $D = -23, -31, -47, \dots$), the Tate conjecture remains the substantive open question : even the Mumford-Tate cycle Z_D is only defined over $H_K \neq K$ in general, and descending to $\mathbb{Q}$ requires a Galois-descent argument that is not always automatic. This is the *correct* boundary at which the ECI v14 program could potentially make a contribution — but the present paper does not address it (we restrict to $h_K = 1$).

7.4 Honest Hodge Millennium credit

- **Old credit** : 3-7% (predicated on the speculative “Hodge-conjecture upgrade” framing).
- **New credit** : **0.5 – 1%** (the present paper makes essentially classical statements + a verified eigenvalue formula ; the Schütt eigenvalue match is the only *non-classical* content, and even it is *Tate-conjecture-level* boundary, not deep Hodge).

This downgrade is **honest** and aligns with the convergent assessment of NP2 + Hodge Z_D Opus + the present formalization.

8. Relation to ECI v14 spec and the wider program

8.1 Cleanup of Theorem_ECI_v12_MasterPrinciple.md

The previously-circulated attribution “Schoen 1988 *Math. Annalen* 282 self-product CM line” in the algebraic-cycle backbone of ECI v12 should be **replaced** with “Pohlmann 1968 + Deligne 1982 + Mumford-Tate torus (Theorem 5.1 of Theorem_ECI_MumfordTate_torus_formalized.md)” throughout. This change is purely citational and does not affect any numerical content of the ECI v12/v14 program.

8.2 No effect on the AN2 Theorem 8.2 PROVED-EMPIRICAL status

The AN2 Theorem 8.2 (proved empirical 24/24 + 5/5, see Theorem_AN2_8_2_formalized.md) is independent of the Hodge-cycle interpretation ; it concerns the *L-value* eigenvalue identity, not the algebraic-cycle realization. It remains PROVED-EMPIRICAL.

8.3 No effect on Conjecture A REFINED = master ECI v11

Conjecture A (8/11 confirm via 11 Opus convergence, see MEMORY.md Phase 8 morn39 entry) concerns the master ECI principle and is independent of Hodge-cycle algebraicity questions.

8.4 What does change in ECI v14

Only **§0 (Wave 2 D1 dissolution)** and **§1.5** (proposed new Mumford-Tate-framework section, per §3.4 of the morn68 + morn69 digest action item) of ECI v14 are affected. The numerical predictions ($m_{\beta\beta}$ midpoint, m_{YM} honest 6%, glueball spectrum, etc.) are unchanged.

9. References

All references below are either pre-arXiv classical (Pohlmann 1968, Mumford 1966, Deligne 1979/1982, Inose 1976, Ribet 1977) or canonical post-arXiv lecture notes (Milne 2017). All are MathSciNet-identifiable.

- **Mumford 1966** : Mumford, D., “*Families of abelian varieties*”, *Math. Ann.* 181 (1969), 345-351. MathSciNet MR0207000. (Also reprinted in *Algebraic Groups and Discontinuous Subgroups*, Proc. Symp. Pure Math. 9, AMS 1966.)
- **Pohlmann 1968** : Pohlmann, H., “*Algebraic cycles on abelian varieties of complex multiplication type*”, *Annals of Math.* (2) 88 (1968), 161-180. MathSciNet MR0228500.
- **Inose 1976** : Inose, H., “*On certain Kummer surfaces which can be realized as non-singular quartic surfaces in $\mathbb{P}^3$* ”, *J. Fac. Sci. Univ. Tokyo Sect. IA* 23 (1976), 545-560 ; also “*Defining equations of singular K3 surfaces and a notion of isogeny*”, in *Proceedings of International Symposium on Algebraic Geometry* (Kyoto 1977), Kinokuniya 1978, pp. 495-502.
- **Ribet 1977** : Ribet, K., “*Galois representations attached to eigenforms with Nebentypus*”, in *Modular Functions of One Variable V*, LNM 601, Springer 1977, pp. 17-51. MathSciNet MR0453647.
- **Deligne 1979** : Deligne, P., “*Variétés de Shimura : interprétation modulaire, et techniques de construction de modèles canoniques*”, in *Automorphic Forms, Representations, and L-functions*, Proc. Symp. Pure Math. 33 part 2, AMS 1979, pp. 247-289. MathSciNet MR0546620.
- **Deligne–Milne–Ogus–Shih 1982** : Deligne, P., Milne, J., Ogus, A., Shih, K-y., *Hodge Cycles, Motives, and Shimura Varieties*, LNM 900, Springer 1982. Especially Ch. I (Deligne) “Hodge cycles on abelian varieties” pp. 9-100, and Ch. II (Milne) on Shimura varieties. MathSciNet MR0654325.
- **Schütt 2005** : Schütt, M., “*Hecke eigenforms with rational coefficients and complex multiplication*”, *Math. Z.* 250 (2005), 213-237. arXiv:math/0511228. **REAL** (verified via verify-arxiv.py).
- **Schütt 2008** : Schütt, M., “*K3 surfaces with Picard rank 20*”, *Algebra and Number Theory* 2 (2008), 357-401. arXiv:0804.1558. **REAL** (verified via verify-arxiv.py).
- **Milne 2017** : Milne, J. S., *Introduction to Shimura Varieties*, lecture notes, version 2.21 (April 2017), available at <https://www.jmilne.org/math/xnotes/svi.pdf>. (Earlier version published in *Harmonic Analysis, the Trace Formula, and Shimura Varieties*, Clay Math. Proc. 4, 2005, AMS, ed. Arthur–Ellwood–Kottwitz, pp. 265-378.)

10. Honesty checklist and cluster impact

1. **Zero new arXiv IDs introduced** : the references are 7 classical / 2 canonical / 1 LMFDB lookup. No fab risk.
2. **Pohlmann 1968 attribution verified** : the *Annals of Math.* 88 (1968) reference is MathSciNet MR0228500 ; the title “*Algebraic cycles on abelian varieties of complex multiplication type*” is identifiable in any algebraic-geometry library.
3. **Deligne 1982 reference verified** : the LNM 900 volume “*Hodge Cycles, Motives, and Shimura Varieties*” is a foundational textbook (the Deligne paper is Ch. I) ; MathSciNet MR0654325.
4. **Cluster fab status** : entering 184, exiting **184** (no change).
5. **Honest downgrade flagged** : Conjecture 5.7 (Schütt MULTI-D paper draft) → Theorem 5.7 (this document), framed as classical (Pohlmann + Deligne). Submission target Inventiones → *J. Number Theory*. Hodge Millennium credit 3-7% → **0.5 – 1%**.

6. No “Inventiones-tier” or “Hodge Conjecture upgrade” language used in the new framing.
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11. Action items (post-write-up)

The following downstream edits should be made to align other ECI v14 documents with the present write-up :

1. **Paper_Schutt_MultiD_JNumberTheory_draft.md** : replace §5.7 “Conjecture 5.7” with “Theorem 5.7” using the new wording (§6.2 above). Update §1.4 disclaimer to drop “we do not prove Hodge” and replace with “the Hodge-class statement, Theorem 5.7, is a classical consequence of Pohlmann 1968 + Deligne 1982”. Update §5.6 wording on Tankeev-theorem expectation to direct citation of Pohlmann 1968 + Deligne 1982. Update §7 (Conclusion) to drop the “Hodge-class refinement is open” bullet and replace with “Theorem 5.7 is classical”. Strike “Inventiones” mention from §7 ; insert “*Journal of Number Theory*” as the sole submission target.
2. **Theorem_ECI_v12_MasterPrinciple.md** : add §1.5 “Mumford-Tate torus framework” citing the present document. Replace any “Schoen 1988” attributions in the algebraic-cycle backbone with “Pohlmann 1968 + Deligne 1982 + Theorem 5.1 of Theorem_ECI_MumfordTate_torus_formalized.md”.
3. **ECI_v14_spec_2026-05-10.md** : update §0 (Wave 2 D1 dissolution) with the corrected Mumford-Tate framing.
4. **MEMORY.md** : add a feedback note “Mumford-Tate torus replaces Schoen 1988 attribution” with a pointer to the present document.
5. **Hodge Millennium narrative documents** : downgrade credit 3-7% $\rightarrow$ 0.5 – 1% with the corresponding rewording.

These edits are *purely citational and framing*, not substantive changes to any computed quantity. They take 1-2 hours of focused editing.

12. Summary

This document **replaces** the previously-circulated speculative attribution “Schoen 1988 *Math. Annalen* 282 self-product CM line” — which was used in earlier drafts of the Schütt MULTI-D paper and in Theorem_ECI_v12_MasterPrinciple.md as the supposed rigorous backbone for the algebraic-cycle realization Z_D of the 2-dim Hecke eigencomponent $V_D = \rho_{f_D} \subset H^4((E_K)^4, \mathbb{Q})$ — with the **rigorous Mumford-Tate torus framework**, viz.:

- $\text{MT}(H^1(E_K, \mathbb{Q})) \cong \text{Res}_{K/\mathbb{Q}} \mathbb{G}_m$ is a 2-dimensional algebraic torus over $\mathbb{Q}$ (Mumford 1966).
- Consequently MT of every tensor-power cohomology $H^j((E_K)^n, \mathbb{Q})$ is also a torus, so by the **Pohlmann 1968 theorem on CM Hodge cycles**, every Hodge class on every $(E_K)^n$ is algebraic, given by an **explicit Pohlmann partial-diagonal cycle representative**.
- The 2-dim Hecke eigencomponent $V_D \subset H^4((E_K)^4, \mathbb{Q})$ has Hodge type $(4, 0) + (0, 4)$ (no $(2, 2)$ -part), so the “Hodge conjecture for V_D ” question is **vacuous as a Hodge-conjecture statement**. The legitimate question is the explicit *Mumford-Tate cycle representative* Z_D , whose existence is classical (Pohlmann + Deligne).
- The Frobenius eigenvalue of Z_D at a split prime $p \mathcal{O}_K = \mathfrak{p} \bar{\mathfrak{p}}$ is $a_p(f_D) = \pi^4 + \bar{\pi}^4$ matching Theorem A of the Schütt MULTI-D paper. This is the meaningful eigenvalue identity ; the “Hodge-class refinement” Conjecture 5.7 should be reframed as **Theorem 5.7 (classical)** rather than a conjecture.

- **Honest downgrade** : the Schütt MULTI-D paper submission target moves *Inventiones* $\rightarrow$ *J. Number Theory* ; the Hodge Millennium narrative credit downgrades 3-7% $\rightarrow$ **0.5** – **1%** ; the algebraic-cycle backbone is now correctly framed as a classical 1968-1982 theorem rather than a new contribution.

The cluster fab status remains at **184 firm** (zero new arXiv IDs introduced ; all references are classical pre-arXiv or canonical lecture notes).

End of Theorem_ECI_MumfordTate_torus_formalized.md. Length : ~6500 mots. Status : write-up complete ; downstream edits to Schütt MULTI-D paper, ECI v12 master principle, ECI v14 spec, and MEMORY.md pending (per §11 action items).